

Milestone 1 Report

Smart MCQ Solver Challenge

NLP Foundation & Semantic Similarity

Objective

The goal of Milestone 1 was to build a foundational understanding of Natural Language Processing (NLP) techniques for multiple-choice question answering. This included performing exploratory data analysis, text preprocessing, TF-IDF-based feature extraction, semantic similarity measurement using cosine similarity, and evaluating performance with the Mean Average Precision at 3 (MAP@3) metric. The milestone establishes a strong baseline for the Smart MCQ Solver Challenge.

Dataset Description

The dataset is from the Smart MCQ Solver Challenge on Kaggle. The training set contains **2,000** multiple-choice questions with the following fields:

- id : Unique identifier
 - prompt : The question text
 - A–E : Five answer options
 - answer : Correct option (ground truth)
-

Tools and Libraries

- Python 3
 - NumPy & Pandas
 - Scikit-learn : `'TfidfVectorizer'`, `'ENGLISH_STOP_WORDS'`, `'cosine_similarity'`
 - String module (for text cleaning)
-

Methodology

1. Exploratory Data Analysis (EDA)

The dataset was loaded and examined for structure, missing values, and basic statistics.

Dataset Summary

- Total records: **2,000**
 - Columns: `id`, `prompt`, `A`, `B`, `C`, `D`, `E`, `answer`
-

2. Distribution of Correct Answers

Option	Frequency
A	369
B	490
C	459
D	358
E	324

Most frequent: Option B (490)

Least frequent: Option E (324)

Therefore,

$$490 + 324 = \mathbf{814}$$

3. Text Preprocessing

- Converted text to lowercase
- Removed punctuation
- Tokenized using whitespace splitting

Vocabulary size from cleaned prompts: **859** unique words.

4. Stop Word Removal

English stop words were removed using
'sklearn.feature_extraction.text.ENGLISH_STOP_WORDS'.

Example (first prompt after cleaning): **13** words remained.

5. TF-IDF Feature Extraction

Combined text ('prompt + A + B + C + D + E') was used to train a
'TfidfVectorizer' (with English stop words removed).

Total features (vocabulary size): **2,762**

6. Cosine Similarity

TF-IDF vectors were generated for the prompt and each option.

Example result (first question vs Option A): **0.2720**

7. Baseline Prediction

For each question:

- Computed TF-IDF vector of the prompt
- Computed TF-IDF vectors of all five options
- Ranked options by cosine similarity
- Selected the highest similarity option as prediction

Accuracy achieved: **13.55%**

8. MAP@3 Evaluation

Implemented Average Precision at K (APK) and Mean Average Precision @ 3.

Example 1

Ground Truth: C

Prediction: C A B $\rightarrow$ MAP@3 = **1.0**

9. Majority Class Baseline

Ranked options by frequency: B > C > A > D > E

Static prediction for all questions: B $\rightarrow$ C $\rightarrow$ A

Overall MAP@3: **0.42125**

10. TF-IDF + Cosine Similarity Pipeline

Full pipeline implemented:

1. Vectorize prompt
2. Vectorize all options
3. Compute cosine similarities
4. Rank options
5. Take top-3 ranked options
6. Evaluate using MAP@3

Final MAP@3 Score: **0.296167**

Key Learnings

- Practical application of core NLP steps: lowercasing, punctuation removal, tokenization, and stop-word filtering.
- Understanding of vocabulary construction and TF-IDF vectorization.
- Using cosine similarity for semantic matching between questions and options.
- Importance of proper evaluation metrics (especially ranking-based MAP@3).
- Comparison between simple baselines (majority class vs similarity-based).
- Building an end-to-end baseline pipeline for MCQ solving.

Conclusion

This milestone successfully established a classical NLP baseline for the Smart MCQ Solver Challenge using TF-IDF and cosine similarity. While the model achieved modest performance (13.55% accuracy and **0.296167** MAP@3), it provides valuable insights and a solid reference point.